

PCB-SAID: A Low-Cost Camera-Based Dataset for Few-Shot SMD Assembly Inspection

Raffaele Mineo^{1, 2*}

raffaele.mineo@unict.it

Amelia Sorrenti^{1, 2*}

amelia.sorrenti@phd.unict.it

Robin Faro¹

robinfaro@deepsensing.it

Gabriele Mineo¹

gabrielemineo@deepsensing.it

Francesco Cancelliere^{1, 2}

francesco.cancelliere@phd.unict.it

Alberto Faro¹

albfaro@deepsensing.it

¹ Deepsensing SRL, Catania, Italy

² University of Catania, Catania, Italy

Abstract

*Automated Optical Inspection (AOI) of surface-mounted device (SMD) assemblies is critical for ensuring quality and yield in electronics manufacturing, yet current systems rely on expensive hardware and handcrafted algorithms that struggle with novel layouts and rare defect modes. We present **PCB-SAID** (Printed Circuit Board – SMD Assembly Inspection Dataset), the first open, few-shot SMD inspection dataset comprising 175 RGB images annotated with 66 fine-grained classes, spanning well-soldered, mis-soldered (misaligned, lifted, rotated), missing, and short-circuit conditions, using a rigorous two-stage expert pipeline (bounding-box localization followed by polygonal mask refinement). We evaluate five state-of-the-art architectures (Fast R-CNN, DETR, YOLOv11, YOLOv12, SAM) under both full-data and few-shot (1, 5 shots) regimes, reporting mAP@[.50:.95], mAP@.50, precision, and recall. Our results reveal optimal performance gaps on rare and subtle anomalies, underscoring challenges in class imbalance and domain shift. The dataset will be made publicly available upon request to support further research in this area.*

Keywords: Printed Circuit Board (PCB), Surface-Mount Device (SMD), Automated Optical Inspection (AOI), Few-Shot Learning, Defect Detection, Semantic Segmentation, Deep Learning, Industrial Computer Vision, Dataset Benchmarking.

1. Introduction

Automated Optical Inspection (AOI) systems serve as the cornerstone of quality control in modern electronics manu-

facturing, ensuring the functional integrity and reliability of printed circuit boards (PCBs) through rapid detection of assembly faults. Traditional AOI solutions rely on specialized hardware - multiple industrial-grade cameras, structured illumination modules, and custom optics - accompanied by rule-based algorithms grounded in geometry-driven models and template matching. Although these systems achieve high throughput and precision (up to 99% detection accuracy and ~5,000 components/hour), they incur substantial capital and maintenance costs, exhibit limited adaptability to new PCB designs, and demand nontrivial configuration overhead requiring expert intervention [11, 17].

In contrast, deep learning-driven, camera-based AOI promises greater flexibility and cost efficiency by exploiting data-driven feature extraction and anomaly detection capabilities of modern neural networks. Recent works have demonstrated that convolutional neural networks (CNNs) and transformer-based architectures can significantly reduce false positives and detect unforeseen defect types compared to traditional rule-based approaches [7, 29]. However, progress in AI-powered AOI is critically hindered by the paucity of high-quality, well-annotated datasets that capture the full spectrum of realistic assembly anomalies - including misalignment, lift, rotation, and short circuits - under few-shot conditions.

Existing PCB inspection corpora illustrate both the potential and the limitations of current open resources:

- **PCB-AOI** (Kaggle): ~1,200 images with bounding-box labels focused on common defect types but lacking fine-grained segmentation and rare assembly errors [16].
- **FICS-PCB / FPIC**: ~9,900 RGB images and over 77,000 component annotations across 31 board types, providing semantic masks under varied lighting but omitting subtle mis-mounted conditions [13, 18].
- **VISION Datasets**: A meta-benchmark of 14 industrial

*These authors contributed equally to this work.

inspection datasets (18,000 images; 44 defect categories) with polygonal annotations, yet limited coverage of SMD assembly anomalies and few-shot scenarios [1].

- **PKU-Market-PCB:** High-resolution board photographs with localized defect regions but confined to a narrow set of components and defect modalities [12].

Despite these efforts, a persistent gap remains: most datasets do not support pixel-level masks for poorly soldered, misaligned, or lifted components, nor are they designed for few-shot learning typical in rapid prototyping or low-volume production.

To address these limitations, we introduce **PCB-SAID** (Printed Circuit Board – SMD Assembly Inspection Dataset), a novel, open-source corpus specifically tailored for few-shot SMD assembly inspection. The dataset features high-resolution RGB imagery captured using commodity cameras, providing a low-cost imaging setup that closely emulates realistic AOI environments. An expert-driven two-stage annotation pipeline ensures comprehensive annotations, producing both bounding-boxes and polygonal masks across 10 common SMD component types and 22 different packaging, in addition to a dedicated short-circuit defect class. The dataset incorporates substantial defect diversity, comprising 66 classes that span well-soldered, missing, and mis-mounted conditions, including shifted, rotated, and lifted assembly faults. These annotations are derived from 175 web-sourced PCB images, reflecting real-world heterogeneity. PCB-SAID has been deliberately designed to support few-shot learning protocols, enabling controlled 1 and 5-shot experiments that facilitate research in meta-learning, domain adaptation, and anomaly detection under severe data scarcity. To establish reference baselines, we report the performance of multiple state-of-the-art architectures, including Fast R-CNN, DETR, YOLOv11, YOLOv12, and SAM, evaluated under both full-data and few-shot regimes using standard metrics such as mAP, mAP@[.50:.95], precision, and recall.

The primary contributions of this paper are therefore threefold:

1. **PCB-SAID Dataset Release:** The first open dataset providing component-level segmentation masks and bounding-boxes for industrial SMD inspection under realistic assembly anomalies;
2. **Annotation Protocol:** A detailed, two-stage expert annotation pipeline ensuring high-fidelity ground truth for both detection and segmentation tasks;
3. **Baseline Evaluations:** Rigorous benchmarks of five state-of-the-art architectures in both full-data and few-shot regimes, accompanied by an analysis of typical failure modes.

2. Background and Related Work

Automated Optical Inspection (AOI) has evolved over three decades as the principal means of detecting assembly defects on printed circuit boards (PCBs). Early AOI systems were conceived around high-resolution imaging hardware and deterministic, rule-based software pipelines. In this section, we first review the classical hardware architectures and illumination strategies, then summarize the core image processing and computer vision algorithms traditionally employed, and finally discuss the transition toward AI-driven approaches.

2.1. AOI Hardware and Illumination

Classical AOI hardware typically comprises one or more industrial-grade CCD or CMOS cameras equipped with telecentric lenses to minimize perspective distortion [11]. Illumination modules generally include structured white light systems, such as multi-angle bar or ring lighting, designed to enhance edge contrast while suppressing specular reflections [5]. Coaxial brightfield and darkfield illumination configurations are also widely adopted, where brightfield illumination aligns with the optical axis to optimize solder joint inspection, and darkfield employs oblique lighting to improve component lead visibility [5]. In more advanced systems, laser triangulation and 3D profilometry are utilized, particularly in 3D-AOI setups, to capture solder paste volume and measure component height variations, thereby improving detection accuracy for lifted or tombstoned components [11]. Although these hardware configurations enable high throughput, often processing thousands of components per hour, they are associated with significant costs, larger physical footprints, and considerable calibration complexity.

2.2. Rule-Based Image Processing Techniques

Traditional AOI software pipelines rely on deterministic image processing sequences applied to the captured imagery. The preprocessing stage typically involves noise reduction using Gaussian or median filtering, contrast enhancement via histogram equalization, and lens distortion correction to rectify optical artifacts [5]. Feature extraction techniques commonly apply edge detection algorithms such as Canny or Sobel operators, alongside morphological filtering like opening and closing operations, to isolate solder blobs and identify relevant structures. Connected-component labeling (blob analysis) further segments individual components [5]. Subsequently, template matching and CAD-to-image comparison methods are employed, where rigid alignment using PCB fiducials allows for pixel-wise correlation between captured regions of interest and CAD-derived templates to detect missing or misaligned components [11]. Geometric and rule-based checks are then applied to verify pad occupancy, lead-to-pad alignment using center-of-mass

displacement thresholds, and to assess solder-joint shape metrics such as area, perimeter, and convexity. Additionally, solder bridges are detected using morphological skeletonization techniques [5].

2.3. Limitations of Conventional AOI

Despite achieving high accuracy under well-controlled production conditions, conventional AOI systems present several limitations. They exhibit high sensitivity to lighting and viewpoint variations, where rule-based thresholds often fail under changing illumination or slight camera pose adjustments, resulting in increased false positive and false negative rates. The configuration process remains highly rigid, as each new PCB design necessitates manual template creation and rule tuning by domain experts. Furthermore, these deterministic algorithms lack generalization capabilities, struggling particularly with novel or subtle defect types such as lifted, rotated, or partially bridged components.

2.4. Emergence of AI-Based AOI

Deep learning approaches have recently been adopted to overcome these limitations. Convolutional Neural Networks (CNNs) and transformer-based detectors (e.g., Fast R-CNN [8], DETR [4]) learn hierarchical feature representations directly from images, obviating the need for hand-crafted rules [21]. Semantic and instance segmentation networks (U-Net [22], Mask R-CNN [10], SAM [15]) further enable precise localization and mask generation for solder joints and component outlines [29]. Despite these advances, the paucity of large, richly annotated AOI datasets constrains model robustness, especially under low-data or few-shot scenarios. Our work on PCB-SAID addresses this gap by providing high-quality, polygonal annotations for a wide spectrum of assembly defects.

2.5. Public Datasets for PCB Inspection

Existing benchmarks for industrial anomaly detection, such as MVTecAD [2] and KolektorSDD [25], have driven progress in unsupervised defect localization by providing high-resolution images of single-component anomalies (e.g., scratches, dents, texture irregularities). However, these datasets focus primarily on surface-level defects and lack multi-class diversity, assembly context, and pixel-accurate masks required for fine-grained printed circuit board (PCB) inspection. PCB-specific collections - such as PKU-Market [12] and the DsPCBSD+ dataset [19] - address certain solder joint anomalies (e.g., insufficient solder, bridging) but often provide only bounding-box annotations, cover a narrow subset of component types (typically resistors and capacitors), and do not encode assembly states (misalignment, lift, rotation). The absence of large, richly annotated PCB datasets with both bounding-box and polygonal segmentation labels hinders evaluation of detec-

tion and segmentation methods under realistic manufacturing scenarios, particularly when classes are highly imbalanced across multiple defect modes.

2.6. Few-Shot and Meta-Learning in Inspection

Few-shot learning (FSL) and meta-learning paradigms aim to overcome data scarcity by enabling rapid adaptation to novel classes with minimal labeled examples. Metric-based approaches - such as Prototypical Networks [23] and Matching Networks [27] - learn embedding spaces in which class prototypes can be computed from few examples, while optimization-based methods like Model-Agnostic Meta-Learning (MAML) [6] tune model parameters for fast fine-tuning on new tasks. In manufacturing vision, FSL has been applied to anomaly detection on MVTecAD and similar datasets [28], demonstrating improved generalization to unseen defect types. Nonetheless, extensions of FSL to dense prediction tasks - such as instance segmentation - or to real-world PCB inspection with dozens of defect categories remain underexplored. Imbalanced class distributions and the need to discriminate subtle geometric perturbations (e.g., component skew, lift) pose additional challenges for conventional FSL frameworks.

2.7. Foundation Models and Promptable Segmentation

The emergence of large, transformer-based foundation models pre-trained on web-scale image corpora offers new avenues for zero-shot and few-shot transfer in industrial inspection. The Segment Anything Model (SAM) [15] exemplifies a promptable segmentation paradigm, capable of generating accurate masks from sparse user prompts (points, boxes, text), thereby drastically reducing annotation overhead. Early studies integrating SAM into manufacturing contexts indicate that prompt-driven segmentation can adapt to new defect modalities with minimal manual intervention [24]. However, the direct application of SAM and similar models to PCB inspection requires careful domain adaptation: color and texture distributions on PCBs differ markedly from natural images, and defect classes can be extremely fine-grained. Hybrid pipelines that combine foundation model proposals with task-specific fine-tuning or metric-learning modules hold promise for scalable, flexible AOI systems capable of handling diverse assembly anomalies across multiple board designs.

3. Data Acquisition and Methodology

3.1. Dataset Collection

PCB-SAID comprises 175 high-resolution RGB images (native resolution 640×480) aggregated from multiple public sources, including electronics enthusiast forums, open-source hardware repositories, and automated web

crawls filtered for PCB content. Each image portrays one or more PCB assemblies under diverse capture conditions:

- **Illumination variation:** ambient fluorescent, directional LED ring, and mixed-shadow settings to emulate real-world shop-floor lighting;
- **Viewpoint perturbations:** slight rotations (up to $\pm 15^\circ$), tilts (up to $\pm 10^\circ$), and scale changes ($0.8\text{--}1.2\times$) reflecting hand-held or low-cost fixture setups;
- **Board diversity:** a variety of form-factors (single-layer to multi-layer), component densities (sparse to high-density), and solder finishes (lead-free, tin-lead).

To ensure representative coverage, we applied a stratified sampling strategy based on board manufacturer and component count distribution, discarding images with excessive glare, motion blur, or occlusions affecting $> 20\%$ of the board area.

Component Type	Well Soldered	Poorly Soldered	Missing
Aluminium Capacitor	14	2	18
Button	2	3	-
Capacitor-Diode-Resistor	-	-	192
Ceramic Capacitor	335	100	-
DO-213AA	1	-	-
Fuse	4	1	-
HC-49S	5	1	-
LED	3	4	-
Mini-USB Socket	3	1	5
PDIP	26	5	17
Power Inductor	7	5	6
QFN-8	-	1	2
Resistor	243	120	-
SMA	5	3	-
SMAF	2	-	-
SMC	5	1	-
SO-13	-	1	-
SO-323	-	2	-
SO-666	-	1	-
SOD-323	3	-	-
SOD-523	16	19	7
SOD-80	2	2	-
SOIC	9	1	7
SOP-16	8	1	-
SOT-223-3L	2	3	-
SOT-23-3L	49	32	33
SOT-323	8	-	2
SOT-343	5	2	1
SOT-353	4	-	-
SOT-666	5	-	-
TSOP	11	-	6
Tantalum Capacitor	19	1	-
Trimmer	3	1	-
Total (SMD)	799	341	296
Short Circuit	56 instances (Solder Bridges)		

Table 1. Aggregated class distribution in PCB-SAID. Each component is annotated with its number of instances across assembly states.

3.2. Annotation Pipeline and Taxonomy

Our annotation process follows a rigorous two-stage protocol to ensure high-fidelity labels for both detection and segmentation. In the first stage, two domain experts independently localized each SMD component and any sol-

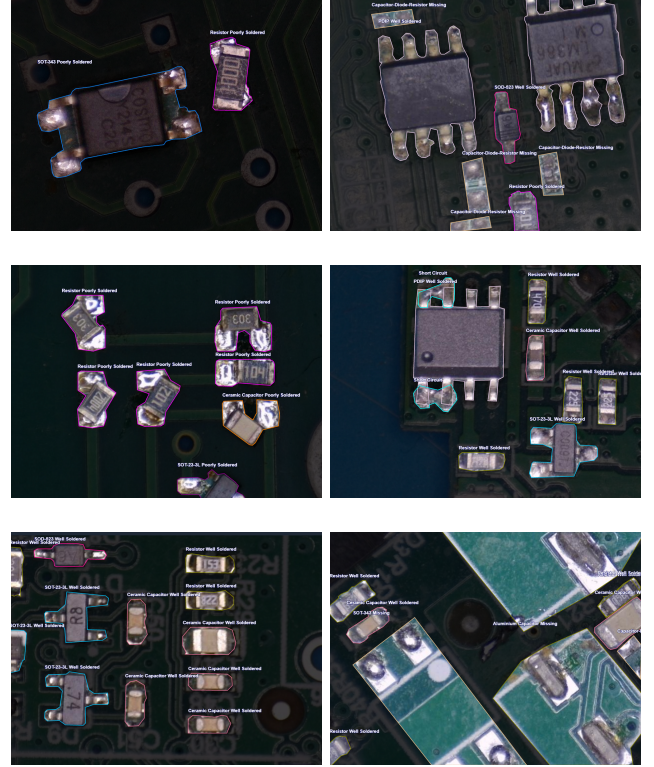


Figure 1. Examples of annotated SMD components in PCB-SAID: well soldered, misaligned, lifted, rotated, missing, and short circuit. The background is blurred only for visualization purposes.

der bridges by drawing axis-aligned bounding boxes using a custom annotation interface. Inter-annotator agreement was assessed via mean Intersection-over-Union (mIoU) and Cohen’s κ , consistently exceeding $\kappa > 0.85$; any bounding-box pairs with $\text{IoU} < 0.75$ were resolved through joint review to guarantee consistency. In the second stage, these bounding boxes were refined into precise polygonal masks: experts traced component perimeters and solder-fillet contours at subpixel precision, aided by a semi-automated contour-fitting module that suggested mask vertices and reduced manual effort by approximately 40%. All final masks were validated for topological consistency (no self-intersections, full closure) and pixel-level accuracy.

We define a fine-grained taxonomy encompassing 66 classes, each corresponding to a unique (*component type*, *assembly state*) pair. Assembly states include well-soldered (proper pad coverage and alignment), poorly soldered (subdivided into misaligned, lifted, and rotated to capture diverse geometric deviations), missing (footprint present without component), and short circuit (solder bridges between adjacent pads or leads). Table 1 provides the per-class instance counts, illustrating the inherent imbalance in real-world PCB layouts (for example, certain poorly sol-

dered or missing conditions appear fewer than five times). To ensure consistent evaluation under few-shot conditions, the dataset is split into training, validation, and test subsets using stratified sampling, such that all 66 classes are represented in each subset with approximately balanced class distributions.

3.3. Data Splits and Pre-processing

For the baseline experiments, the dataset was partitioned into training, validation, and test subsets following a 70%-15%-15% split, ensuring that no PCB image appears in more than one subset to prevent data leakage. Each original training image was further expanded by generating augmented versions, applying a randomized sequence of transformations designed to simulate real-world variability encountered in practical AOI scenarios.

The augmentation pipeline included horizontal and vertical flipping, as well as random 90° rotations, with the rotation angle selected uniformly among 90° clockwise, 90° counter-clockwise, and 180°. Random cropping was applied by uniformly sampling a zoom factor between 100% (no zoom) and 50% (maximum zoom-in). Arbitrary in-plane rotation was introduced by sampling the rotation angle from the interval $[-15^\circ, +15^\circ]$, while shear transformations were independently applied in both horizontal and vertical axes, with angles drawn from a uniform distribution within $\pm 20^\circ$. Grayscale conversion was applied to 31% of the augmented samples to simulate variations in sensor color fidelity.

Further photometric augmentations included hue adjustments sampled from $[-80^\circ, +80^\circ]$, saturation shifts within $[-50\%, +50\%]$, brightness modifications in the range $[-50\%, +50\%]$, and exposure changes between $[-30\%, +30\%]$. To emulate optical blur and sensor imperfections, Gaussian or motion blur with kernel sizes up to 10 pixels was applied, along with salt-and-pepper noise affecting up to 0.1% of image pixels. Finally, cutout augmentation was employed by randomly inserting five square masks per image, each covering approximately 10% of the image area.

Each augmentation parameter was sampled independently for every image and applied in randomized order, ensuring maximal diversity in the generated training samples. This comprehensive preprocessing and augmentation strategy was consistently employed across all baseline and few-shot experiments detailed in Section 3.4.

3.4. Baseline Model Training

We establish a comprehensive benchmark on PCB-SAID using five state-of-the-art detection and segmentation architectures. All models are fine-tuned on 70% of the dataset (train split) with the augmentation pipeline described in Section 3. Hyperparameters are selected via grid search on

the 15% validation split, and training is terminated when validation loss plateaus for 10 consecutive epochs.

- **Fast R-CNN** [8]: Two-stage detector with a ResNet-50 [9] backbone pre-trained on ImageNet. We replace the RoI head to predict 66 classes and fine-tune with Adam (learning rate 5e-4, weight decay 1e-4) for a total of 113 epochs;
- **DETR** [4]: End-to-end transformer detector with a ResNet-50 CNN backbone. Optimization uses AdamW (learning rate 1e-4, weight decay 1e-4) for 71 epochs;
- **YOLOv11** [14]: Single-shot detector with CSPDarknet53 [20] backbone. Trained using Adam (lr 1e-3, weight decay 1e-4) for 188 epochs and multi-scale input augmentation (image sizes between 416 and 608 px);
- **YOLOv12** [26]: Improved variant of YOLOv11 featuring BiFPN feature fusion. Same optimizer settings as YOLOv11, trained for 164 epochs with extended warm-up;
- **SAM** [15]: Promptable segmentation model. We generate mask proposals using 5 center prompts per component and fine-tune the mask head with Adam (lr 5e-5) for 56 epochs, keeping the image encoder frozen.

3.5. Evaluation Metrics

We evaluate each model on the held-out 15% test split using the following metrics:

- **mAP@[.50:.95]**: mean average precision over IoU thresholds from 0.50 to 0.95 in steps of 0.05;
- **mAP@.50**: average precision at IoU = 0.50;
- **Precision**: class-averaged precision at detection threshold;
- **Recall**: class-averaged recall at detection threshold.

Table 2 summarizes the quantitative results.

Model	mAP@.50	Precision	Recall
Fast R-CNN	71.9	61.9	85.4
DETR	74.9	66.5	80.0
YOLOv11	73.1	64.7	76.8
YOLOv12	76.3	67.5	70.4
SAM	67.2	54.1	72.6

Table 2. Baseline performance on PCB-SAID test split.

3.6. Few-Shot Learning Experiments

To evaluate model adaptability under severe data scarcity, we perform few-shot experiments across the full set of 66 classes defined in Section 3. For each shot level ($k \in \{1, 5\}$), we randomly sample k labeled instances per class from the training split and fine-tune each backbone initialized from its full-data checkpoint. We implement a

prototype-based [23] adaptation strategy: class prototypes are computed in the feature embedding space (extracted from the penultimate network layer), and the classifier head is fine-tuned using a cosine similarity loss between embedded queries and class prototypes. Performance is reported in terms of mAP@[.50:.95] for each shot level in Table 3.

Model	1-shot	5-shot
Fast R-CNN	18.3	37.6
DETR	21.7	40.9
YOLOv11	21.1	36.4
YOLOv12	26.8	46.4

Table 3. Few-shot performance (mAP@[.50:.95]) on novel classes.

4. Conclusion and Future Work

In this paper, we have introduced PCB-SAID (Printed Circuit Board – SMD Assembly Inspection Dataset), a novel open dataset designed for few-shot SMD assembly inspection. PCB-SAID comprises 175 high-resolution RGB images, each annotated with both axis-aligned bounding boxes and high-precision polygonal masks across 66 fine-grained defect-oriented classes. We presented a rigorous two-stage annotation pipeline, outlined our preprocessing and comprehensive augmentation strategy, and established baseline benchmarks on five state-of-the-art architectures (Fast R-CNN, DETR, YOLOv11, YOLOv12, and SAM) under both full-data and few-shot regimes. Our experimental results highlight the capabilities and limitations of current detection and segmentation models when confronted with subtle assembly anomalies (such as misalignment, lift, rotation, and solder bridges) and expose significant performance gaps on rare defect classes.

Despite these contributions, key challenges remain. First, class imbalance and the scarcity of rare defects (e.g., lifted SO/SOD or skewed QFN packages) constrain model exposure during training and limit generalization to infrequent failure modes. Second, domain shifts arising from variations in lighting conditions, board materials, and imaging hardware across manufacturing sites can degrade cross-site performance. Third, real-time deployment requirements on edge-mounted GPUs or embedded vision systems impose stringent constraints on model latency and memory footprint.

To address these limitations, we propose several promising directions for future research. On the data front, large-scale acquisition of PCB images - via collaboration with industry partners - can extend coverage to multilayer, flexible, and fine-pitch assemblies, while synthetic data gen-

eration through physics-based rendering engines can inject parametrizable defects at scale.

On the modeling side, self-supervised pretraining on large-scale unlabeled PCB corpora - using contrastive learning or masked image modeling - can yield task-agnostic representations that accelerate downstream defect detection. Adaptive fine-tuning of promptable segmentation models (e.g., SAM) with lightweight adapters or LoRA modules promises rapid domain adaptation with minimal annotation overhead. Hierarchical meta-learning frameworks that jointly optimize detection, segmentation, and classification objectives may further enhance generalization across evolving component libraries. Finally, neural architecture search and model quantization tailored for industrial edge devices will be critical to achieving sub-50 ms inference times while maintaining high accuracy.

Ultimately, we anticipate that integrating PCB-SAID-trained models into closed-loop manufacturing systems - where inspection feedback dynamically informs placement operations [3], soldering parameters, and in-line fault recovery - will realize a new generation of intelligent, resilient AOI solutions. Such systems will require seamless interoperability with manufacturing execution systems (MES), real-time anomaly reporting, and explainable AI modules capable of diagnosing root causes. By advancing both dataset richness and model sophistication, we aim to catalyze research toward flexible, efficient, and scalable inspection frameworks for next-generation electronics manufacturing.

5. Acknowledgments

This research was conducted as part of the EdgeAI "Edge AI Technologies for Optimised Performance Embedded Processing" project, which has received funding from KDT JU under grant agreement No 101097300. The KDT JU receives support from the European Union's Horizon Europe research and innovation program and Austria, Belgium, France, Greece, Italy, Latvia, Luxembourg, Netherlands, and Norway. The work of Raffaele Mineo has been supported by EC Horizon Program (HORIZON-KDT-JU-2022-2-RIA-Topic-1), under project Neurokit2E (No. 101112268).

References

- [1] Haoping Bai, Shancong Mou, Tatiana Likhomanenko, Ramazan Gokberk Cinbis, Oncel Tuzel, Ping Huang, Jiulong Shan, Jianjun Shi, and Meng Cao. Vision datasets: A benchmark for vision-based industrial inspection. *arXiv preprint arXiv:2306.07890*, 2023. 2
- [2] Paul Bergmann, Michael Fauser, David Sattlegger, and Carsten Steger. Mvtec ad—a comprehensive real-world dataset for unsupervised anomaly detection. In *Proceedings*

- of the IEEE/CVF conference on computer vision and pattern recognition, pages 9592–9600, 2019. 3
- [3] Claudio Cantone and Alberto Faro. An overview of the automated optical inspection edge ai inference system solutions. *Advancing Edge Artificial Intelligence*, page 153, 2024. 6
 - [4] Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In *European conference on computer vision*, pages 213–229. Springer, 2020. 3, 5
 - [5] Yajun Chen, Yuanyuan Ding, Fan Zhao, Erhu Zhang, Zhangnan Wu, and Linhao Shao. Surface defect detection methods for industrial products: A review. *Applied Sciences*, 11(16): 7657, 2021. 2, 3
 - [6] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning*, pages 1126–1135. PMLR, 2017. 3
 - [7] Harshitkumar Ghelani. Ai-driven quality control in pcb manufacturing: Enhancing production efficiency and precision. *Valley International Journal Digital Library*, pages 1549–1564, 2024. 1
 - [8] Ross Girshick. Fast r-cnn. In *International Conference on Computer Vision (ICCV)*, 2015. 3, 5
 - [9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. 10.48550. *arXiv preprint arXiv:1512.03385*, 2015. 5
 - [10] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision*, pages 2961–2969, 2017. 3
 - [11] IPC: Chino Hills. Ipc-a-610; acceptability of electronic assemblies. *IPC Standards*, 2017. 1, 2
 - [12] Weibo Huang and Peng Wei. A pcb dataset for defects detection and classification. *arXiv preprint arXiv:1901.08204*, 2019. 2, 3
 - [13] Nathan Jessurun, Olivia P Dizon-Paradis, Jacob Harrison, Shajib Ghosh, Mark M Tehranipoor, Damon L Woodard, and Navid Asadizanjani. Fpic: a novel semantic dataset for optical pcb assurance. *ACM Journal on Emerging Technologies in Computing Systems*, 19(2):1–21, 2023. 1
 - [14] Rahima Khanam and Muhammad Hussain. Yolov11: An overview of the key architectural enhancements. *arXiv preprint arXiv:2410.17725*, 2024. 5
 - [15] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4015–4026, 2023. 3, 5
 - [16] Dongdong Li, Dan Liu, Yun Shen, Yaqi Song, and Lianliang Luo. Pcb-aoi, 2022. 1
 - [17] Qin Ling and Nor Ashidi Mat Isa. Printed circuit board defect detection methods based on image processing, machine learning and deep learning: A survey. *IEEE access*, 11: 15921–15944, 2023. 1
 - [18] Hangwei Lu, Dhvani Mehta, Olivia Paradis, Navid Asadizanjani, Mark Tehranipoor, and Damon L Woodard. Fics-pcb: A multi-modal image dataset for automated printed circuit board visual inspection. *Cryptology ePrint Archive*, 2020. 1
 - [19] Shengping Lv, Bin Ouyang, Zhihua Deng, Tairan Liang, Shixin Jiang, Kaibin Zhang, Jianyu Chen, and Zhuohui Li. A dataset for deep learning based detection of printed circuit board surface defect. *Scientific Data*, 11(1):811, 2024. 3
 - [20] Marsa Mahasin and Irma Amelia Dewi. Comparison of csp-darknet53, espresnext-50, and efficientnet-b0 backbones on yolo v4 as object detector. *International journal of engineering, science and information technology*, 2(3):64–72, 2022. 5
 - [21] Inc. Neurala. Automating visual inspections for manufacturing with deep learning. 2020. 3
 - [22] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5–9, 2015, proceedings, part III 18*, pages 234–241. Springer, 2015. 3
 - [23] Jake Snell, Kevin Swersky, and Richard Zemel. Prototypical networks for few-shot learning. *Advances in neural information processing systems*, 30, 2017. 3, 6
 - [24] Kechen Song, Wenqi Cui, Han Yu, Xingjie Li, and Yunhui Yan. Samera: Can it segment any industrial surface defects? *Computers, Materials & Continua*, 78(3), 2024. 3
 - [25] Domen Tabernik, Samo Šela, Jure Skvarč, and Danijel Skočaj. Segmentation-Based Deep-Learning Approach for Surface-Defect Detection. *Journal of Intelligent Manufacturing*, 2019. 3
 - [26] Yunjie Tian, Qixiang Ye, and David Doermann. Yolov12: Attention-centric real-time object detectors. *arXiv preprint arXiv:2502.12524*, 2025. 5
 - [27] Oriol Vinyals, Charles Blundell, Timothy Lillicrap, Daan Wierstra, et al. Matching networks for one shot learning. *Advances in neural information processing systems*, 29, 2016. 3
 - [28] Shenxing Wei, Xing Wei, Zhiheng Ma, Songlin Dong, Shaochen Zhang, and Yihong Gong. Few-shot online anomaly detection and segmentation. *Knowledge-Based Systems*, 300:112168, 2024. 3
 - [29] Qianru Zhang, Meng Zhang, Chinthaka Gamanayake, Chau Yuen, Zehao Geng, Hirunima Jayasekara, Chia-wei Woo, Jenny Low, Xiang Liu, and Yong Liang Guan. Deep learning based solder joint defect detection on industrial printed circuit board x-ray images. *Complex & Intelligent Systems*, 8(2):1525–1537, 2022. 1, 3